Networking Fundamentals

A computer network is a group of connected devices that exchange data and share resources. A Local Area Network (LAN) covers a limited area such as a home, office, or campus. A Wide Area Network (WAN) spans a larger geographic area and can connect multiple LANs.

A network switch connects devices within a LAN. It forwards Ethernet frames using the destination device's MAC address. A router connects different networks and forwards IP packets using IP addresses and routing information.

An IPv4 address is a 32-bit address commonly written as four decimal octets, for example 192.168.1.10. A subnet mask identifies which part of an IPv4 address refers to the network and which part refers to a host. Devices in different IP networks generally need a router to communicate.

The Domain Name System (DNS) translates human-readable domain names into IP addresses used by networked systems. The Dynamic Host Configuration Protocol (DHCP) can automatically provide a host with configuration such as an IP address, subnet mask, default gateway, and DNS server.

A default gateway is the router interface a host uses to send traffic to destinations outside its local IP network. If a host only needs to communicate with another host on the same IP network, it can send directly on the local network after resolving the destination's link-layer address.

Transport and Network Security

The Transmission Control Protocol (TCP) provides a reliable, ordered byte stream between applications. It uses acknowledgements and retransmission to help deliver data. The User Datagram Protocol (UDP) sends datagrams without TCP's built-in delivery and ordering guarantees. Applications choose a transport protocol based on their requirements.

A port number helps identify an application endpoint on a host. IP addressing identifies hosts and routes packets between networks; transport-layer ports help deliver data to the intended application. A protocol's well-known port is a convention, not proof that a service is actually running there.

The Open Systems Interconnection (OSI) model describes network communication in seven layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application. It is a conceptual model used to discuss and troubleshoot networking; real protocol suites do not always map perfectly to its layers.

A firewall applies rules to network traffic. Depending on its type and configuration, it may allow or block traffic based on attributes such as addresses, ports, protocols, connection state, or application identity. A firewall should be configured according to the network's security requirements.

When troubleshooting, identify the scope of the problem first. Check physical connectivity, IP configuration, local network reachability, default gateway reachability, and name resolution as appropriate. A successful DNS lookup does not by itself prove that the destination application is available.